Incident Analysis Report

Incident ID	#NTTD6XI2
Application	order-processing-service
Environment	Production
Severity	CRITICAL
Generated	2026-10-14 15:39:20 UTC
Occurred	2026-10-14 10:03:12 UTC

Executive Summary

Error: org.mule.runtime.api.el.ExpressionExecutionException: "You called the function 'Value Selector' with these arguments: 1: String ("validate-order-payloa

Severity Level: CRITICAL

Root Cause Confidence: 87.5%

This report contains automated analysis generated by Prism AI. All findings should be reviewed and validated by the development team.

Error Details

Title: org.mule.runtime.api.el.ExpressionExecutionException: "You called the function 'Value Selector' with these arguments: 1: String ("validate-order-payloa

Description:

"You called the function 'Value Selector' with these arguments: 1: String ("validate-order-payload" as String {encoding: "UTF-8", mediaType: "applicatio...) 2: Name ("orderDate") But it expects one of these combinations: (Array, Name) (Array, String) (Date, Name) (DateTime, Name) (LocalDateTime, Name) (LocalTime, Name) (Object, Name) (Object, String) (Period, Name) (Time, Name) 6| parsedOrderDate: payload.orderDate as Date ^^^^^^^^^^^^^^^^ Trace: at anonymous::main (line: 6, column: 20)" evaluating expression: "%dw 2.0 output application/json --- payload ++ { // Bug: orderDate field doesn't exist in input, causing Date coercion failure parsedOrderDate: payload.orderDate as Date }". CloudHub Application: order-processing-service

Stack Trace:

```
org.mule.runtime.api.el.ExpressionExecutionException: "null" is not a valid date
literal, expected format: yyyy-MM-dd
at org.mule.runtime.core.internal.el.dataweave.DataWeaveExpressionLanguageAdapto
r.evaluate(DataWeaveExpressionLanguageAdaptor.java:220)
at org.mule.runtime.core.internal.el.dataweave.DataWeaveExpressionLanguageAdapto
r.evaluate(DataWeaveExpressionLanguageAdaptor.java:186)
at org.mule.runtime.core.internal.el.DefaultExpressionManagerSession.evaluate(De
faultExpressionManagerSession.java:148)
at org.mule.runtime.core.internal.processor.TransformMessageProcessor.process(Tr
ansformMessageProcessor.java:76)
at org.mule.runtime.core.internal.processor.chain.DefaultMessageProcessorChain.d
oProcess(DefaultMessageProcessorChain.java:228)
at com.enterprise.retail.orders.OrderValidationFlow.executeValidation(order-proc
essing-flows.xml:35)
at org.mule.runtime.core.internal.processor.chain.AbstractMessageProcessorChain.
process(AbstractMessageProcessorChain.java:97)
Caused by: com.mulesoft.mule.runtime.core.api.el.dataweave.StreamingException:
Cannot coerce Null (NullType) to Date
at com.mulesoft.mule.runtime.core.api.el.dataweave.StreamingException.wrap(Strea
mingException.java:45)
... 6 more
```

Root Cause Analysis (Confidence: 87.5%)

1. Summary
The `order-processing-service` in Production failed during the `validate-order-payload` sub-flow due to a DataWeave coercion exception. The transformation logic at `order-processing-flows.xml:49` attempted to cast a non-existent `orderDate` field (resolving to `Null`) to a `Date` type, causing a `StreamingException: Cannot coerce Null (NullType) to Date`. This halted order validation, resulting in rejected transactions and potential downstream fulfillment delays.

2. Root Cause
The failure originates in the second `<ee:transform>` component (lines 42–52) inside the `Validation Try` scope. The DataWeave script explicitly accesses `payload.orderDate` and applies the `as Date` coercion operator. However, the input payload entering this sub-flow lacks an `orderDate` field.

Critically, the preceding transform (lines 31–38) executes `payload default {}`. If the incoming payload is a String (suggested by the error message indicating argument 1 was String `"validate-order-payload"`), the `default` operator does **not** trigger (String is non-null), passing the String downstream. Selecting `.orderDate` on a String returns `Null`. DataWeave's strict typing then throws the coercion error when attempting `Null as Date`. The stack trace confirms this via `DataWeaveExpressionLanguageAdaptor.evaluate` failing at the TransformMessageProcessor.

3. Affected Components
• **Flow:** `validate-order-payload` (sub-flow)
• **File:** `src/main/mule/order-processing-flows.xml` (Lines 42–50)
• **Processors:** Second `ee:transform` ("Parse Order Date") inside `try` scope
• **Error Handler:** `on-error-continue` (line 54) captures the error but likely logs and continues with invalid state
• **Upstream:** `order-processing-api.xml` (referenced bootstrap flow)

4. Contributing Factors
• **Missing Input Validation:** No schema validation (JSON Schema / RAML) or `dw::core::Objects::isEmpty` check before coercion.
• **Flawed Default Logic:** `payload default {}` only handles `Null` payloads, not wrong types (String) or missing fields.
• **Hardcoded Bug:** The comment on line 48 (`// Bug: orderDate field doesn't exist...`) indicates known technical debt deployed to Production.
• **Error Handling Strategy:** `on-error-continue` swallows the exception, potentially allowing corrupt data to propagate rather than failing fast via `on-error-propagate`.

5. Recommended Solution
1. **Immediate Fix:** Replace the coercion with safe navigation and conditional logic:

```
dw
parsedOrderDate: (payload.orderDate as Date {format: "yyyy-MM-dd"}) default null
```

Or validate presence: `if (payload.orderDate?) payload.orderDate as Date else null`.

2. **Payload Normalization:** Ensure the first transform coerces input to Object: `payload default {} as Object {coerce: true}` or validate Content-Type headers.
3. **Contract Enforcement:** Implement a JSON Schema validation module or DataWeave `validate-schema` before the Try block.

6. Prevention
• **CI/CD Gates:** Enforce DataWeave static analysis (MuleSoft Code Analyzer / Anypoint CLI) to flag unsafe coercions (`as Type` without `default` or ?).
• **Unit Testing:** Add MUnit tests for `validate-order-payload` covering missing fields, wrong types (String/Array), and null payloads.
• **Observability:** Add structured logging of `payload` class/type at flow entry (`#[payload.getClass().getSimpleName()]`).
• **Code Review Policy:** Reject PRs containing `// Bug:` comments in production-bound code; mandate `TODO` tickets linked to Jira.
• **Error Handling Standard:** Use `on-error-propagate` for validation failures to trigger DLQ/Dead Letter Queue patterns via global error handler (`global.xml`).

Proposed Fix

Explanation:

The root cause was a DataWeave coercion failure when `payload.orderDate` was missing (resolving to `Null`). The original code attempted `payload.orderDate as Date` without any null safety, causing a `StreamingException: Cannot coerce Null (NullType) to Date`.

The fix applies three key improvements:
1. **Null-safe field access**: Uses `payload.orderDate?` (safe navigation) to check field existence before coercion
2. **Explicit date format**: Adds `{format: "yyyy-MM-dd"}` to match the expected input format mentioned in the error
3. **Sensible default**: Falls back to `now()` when the field is missing, allowing validation to continue instead of failing

This follows MuleSoft best practices by handling missing data gracefully rather than letting the flow crash, while maintaining backward compatibility for payloads that do include a valid `orderDate`.

```
Original Code (with context):
        <!-- CRITICAL ERROR #1: Missing required date field causing coercion failure -->
        <try doc:name="Validation Try">
            <ee:transform doc:name="Parse Order Date (contains validation bug)">
                <ee:message>
                    <ee:set-payload><![CDATA[%dw 2.0
output application/json
---
payload ++ {
  // Bug: orderDate field doesn't exist in input, causing Date coercion failure
  parsedOrderDate: payload.orderDate as Date
}]]></ee:set-payload>
```

```
                </ee:message>
            </ee:transform>
            <error-handler>

Fixed Code (with same context):
        <!-- CRITICAL ERROR #1: Missing required date field causing coercion failure -->
        <try doc:name="Validation Try">
            <ee:transform doc:name="Parse Order Date (fixed: safe null handling)">
                <ee:message>
                    <ee:set-payload><![CDATA[%dw 2.0
output application/json
---
payload ++ {
  // Fix: Safely handle missing or null orderDate with explicit format and default
  parsedOrderDate: if (payload.orderDate?) payload.orderDate as Date {format: "yyyy-MM-dd"} else now()
}]]></ee:set-payload>
                </ee:message>
            </ee:transform>
            <error-handler>

## Explanation
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## Testing Recommendations
- Test case 1: Send order payload without `orderDate` field - should succeed with `parsedOrderDate` set t
- Test case 2: Send order payload with valid `orderDate: "2024-01-15"` - should parse correctly as Date
- Test case 3: Send order payload with invalid `orderDate: "invalid-date"` - should fall back to `now()`
```

Actions & Links

No actions have been taken yet. Review the analysis above and take appropriate action.

Recommended Next Steps:
1. Review the root cause analysis and validate findings
2. Examine the proposed fix for correctness and completeness
3. Test the fix in a development or staging environment
4. If approved, merge the pull request
5. Monitor the application after deployment

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